Original Article

Intestinal Fatty Acid-binding Protein in Patients with Psoriasis: A Marker of Impaired Intestinal Permeability Barrier and its Relation to Demographic Data and Clinical Characteristics

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Abstract

Background: Alterations of intestinal microbiota may play a role in the pathogenesis of psoriasis. Dysbiosis may cause disruption of the intestinal barrier, which contributes to immune activation by translocation of microbial antigens and metabolites. Intestinal fatty acid-binding protein (I-FABP) serves as a biomarker of enterocyte damage. **Purpose:** The aim of this study was to investigate serum concentration of I-FABP in patients with psoriasis. **Methods:** A cross-sectional hospital-based study on fifty psoriatic patients and thirty-five age and sex-matched healthy volunteers as a control group were enrolled in the study, serum I-FABP concentration was measured using an enzyme-linked immunosorbent assay. **Results:** Concentration of serum I-FABP was higher in patients compared to controls ($P = 0.04$). The serum level of I-FABP was higher in patients with skin phototype IV than in those with skin phototype III ($P = 0.040$). There were significant positive statistical correlations between I-FABP with age and disease duration. **Conclusion:** I-FABP, a biomarker for gut permeability, is increased in psoriasis and correlates with disease duration and age. Further investigations are needed to determine whether reinforcing intestinal barrier may be a new therapeutic target in psoriasis.

Keywords: Intestinal barrier integrity, intestinal fatty acid-binding protein, psoriasis

Introduction

Psoriasis is a common chronic immune-mediated skin disorder that causes significant physical and social distress.[1] Its prevalence ranges from 0.91% to 8.5% in adults and 0%–2.1% in children.[2] The exact etiology of psoriasis is unsettled. However, the primary cause is mostly a dysregulated immune response, which manifests in people who have one or more psoriasis susceptibility genes after exposure to a triggering factor. Pathophysiologic events include epidermal hyperproliferation, abnormal keratinocytes' differentiation, and inflammation, as well as cutaneous immunologic changes.[3]

Recently, considerable interest has been focused on the possible interaction between gut microbiota, the intestinal barrier and the immune system. Intestinal barrier dysfunction is a consequence of many factors, including gut dysbiosis. The dysfunctional intestinal barrier may be a factor in the etiopathogenesis of psoriasis and other skin diseases with potentially attractive therapeutic implications.[4]

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The gut microbiota is a complex community of microorganisms that includes about 1,000 different species of bacteria, viruses, fungus, and protozoa.[5] Gut commensals primarily aid in nutrition metabolism, medication metabolism, pathogen colonization prevention, and intestinal barrier integrity.[6] Dysbiosis, or changes in the microbiota's composition and function, has been linked to the onset and progression of several skin diseases, including psoriasis. The specific mechanism of this crosstalk has been suggested to be due to increased gut barrier permeability.[7]

The intestinal mucosa comprises a mechanical, chemical, biological and immune barrier. It is the largest surface

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for interaction with microorganisms in the human body. Disruption of the intestinal barrier can cause translocation of bacteria, toxins and metabolites into the circulation which further induces or aggravates systemic inflammation and immunological dysfunction.[8]

A lining of enterocytes and tight connections maintains the intestinal barrier function by sealing the paracellular gap between adjacent enterocytes. Intestinal barrier integrity can be evaluated by assessment of intestinal epithelial cell injury or tight junction loss.[9]

Intestinal fatty acid-binding protein (I-FABP) is a 15-kDa cytosolic protein. It is involved in the uptake and trafficking of fatty acids in the small and large intestines. This protein is mainly expressed by enterocytes at the tips of intestinal villi. It is released when the cell membrane integrity is compromised and has been used as a serum biomarker for the dysfunctional intestinal barrier.[10] I-FABPs are tiny, water-soluble cytosolic proteins that are easily secreted into the circulation and filtered by the kidneys when enterocyte membrane integrity is lost.[9] So, the I-FABPs level is reliable for early diagnosis of intestinal epithelial cell injury because they can be detected sensitively in both plasma and urine using enzyme-linked immunosorbent assay (ELISA).[11]

The aim of the work is to assess I-FABP as a noninvasive marker of intestinal barrier integrity in patients with psoriasis.

Methods

A case–control study was carried out in Dermatology, Venereology and Andrology Department in collaboration with the Clinical Pathology Department, Assiut University Hospitals, Assiut, Egypt. The sample size was calculated using G power software version 3.1.3, Heinrich-Heine-University, Düsseldorf, Germany. To detect a significant difference in the mean of serum I-FABP between two studied independent groups (psoriasis cases and control group) and by applying the following parameters: *t*-test, effect size 1.46, alpha error 0.01, power 0.95, and allocation ratio 1:1. The minimum required sample size was 38 subjects (19 subjects in each group). This study included fifty psoriatic patients and thirty-five age and sex-matched healthy volunteers as a control group.

Exclusion criteria included

age below 18 years, history of gastrointestinal tract infection or surgery during the last 3 months before the study, dietary restriction during the last 3 months, intake of probiotics or prebiotics during the last 1 month, use of topical therapy during last month, systemic anti-psoriatic treatment or phototherapy in the previous 3 months, pregnancy or breastfeeding and any associated skin or systemic illness.

Detailed medical history, general and dermatologic examinations, and psoriasis severity by psoriasis area severity index (PASI) were assessed.[12] The serum concentration of I-FABP was measured by ELISA according to the manufacturer's instructions (SinoGeneClon Biotech Co., China) The Committee of Medical Ethics, Faculty of Medicine, Assiut University, Egypt reviewed and approved this research (IRB no. 17300348). ClinicalTrials.gov ID: NCT04212507.

Data entry and analysis were done using SPSS (Statistical Package for the Social Science) version 19 (SPSS Inc., Chicago, IL, USA). Quantitative data were expressed as mean ± standard deviation (SD), median and interquartile range. Qualitative data were expressed as numbers and percentages (No, %). The Chi-square test and Fisher's exact test were used to compare the categorical variables. Comparisons between continuous variables were done by *t*-test or Mann–Whitney test. Pearson correlation or Spearman's coefficient was used to assess the associations among continuous variables. *P* value was considered statistically significant when $P < 0.05$.

Results

This study included 50 patients with different clinical types of psoriasis including 29 (58.0%) males and 21 (42.0%) females. Their ages ranged from 18 to 80 years. Thirty-five healthy matched volunteers were included as controls [Table 1]. Nearly all subjects had pruritus [Table 2]. Twenty-nine patients (58%) had skin type IV, and 21 (42%) had skin type III. Plaque psoriasis was present in 45 (90%) patients, erythrodermic psoriasis in 2 (4%) patients, and guttate, pustular, and inverse psoriasis were seen in one patient (2%), each. Nail involvement was present in 27 patients (54%). The scalp was affected in 23 patients (46%). The range of PASI score was (0.3–64) with a mean ± SD of 10.75 ± 11.34, a median of 6.85 and an interquartile range of 12.75. Stress was the main triggering factor of psoriasis in 28 patients (56%), change of season (cold weather) in 27 patients (54%), trauma in 24 patients (48%) and smoking in one patient (2%).

The level of I-FABP in the blood was higher in psoriatic patients than in controls [$P = 0.04$, Table 3], but there was no correlation between I-FABP with the PASI score [Table 4]. The serum level of I-FABP was higher in patients with skin phototype IV than in patients with phototype III ($P = 0.04$), with no significant relation between I-FABP levels with gender, family history or clinical type [Table 5]. I-FABP level

Table 1: Demographic data in patients and controls

Characteristics	Patients (*n*=50)	Controls (*n*=35)	*P*
Gender, *n* (%)			
Male	29 (58.00)	20 (57.10)	0.93
Female	21 (42.00)	15 (42.90)	
Age (years)			
Mean±SD	45.26±15.9	42.28±14.70	0.46
Range	18–80	18–77	
Median	45.50	42.00	
IQR	24.75	23.00	

No statistically significant difference ($P>0.05$). IQR: Interquartile range, SD: Standard deviation

Table 2: Clinical data of patients

	Value
Age of onset	
Mean±SD	32.46±16.80
Range (years)	1–67
Median	30.00
IQR	29.50
Disease duration	
Mean±SD	12.82±12.67
Range (years)	0.5–49
Median	8.50
IQR	14.75
Course, *n* (%)	
Progressive	34 (68)
Remission and exacerbation	10 (20)
Regressive	4 (8)
Stationary	2 (4)
Pruritus, *n* (%)	
Yes, only on the lesion	47 (94)
Yes, all over the body	1 (2)
No	2 (4)
Family history of psoriasis, *n* (%)	
Yes	15 (30)
No	35 (70)

IQR: Interquartile range, SD: Standard deviation

Table 3: The levels intestinal fatty acid binding protein in patients with psoriasis and controls

	I-FABP (ng/ml)		*P*
	Patients	Controls	
Minimum	0.10	1.10	0.04*
Maximum	55.30	9.50	
Mean±SD	10.14±13.53	4.50±2.26	
Median	5.75	4.40	
IQR	3.70	4.00	

*Statistically significant difference (P<0.05). IQR: Interquartile range, I-FABP: Intestinal fatty acid binding protein, SD: Standard deviation

was correlated with age patients› age (P = 0.025) and disease duration (P = 0.036). Surprisingly, we could not detect any significant.

Table 4: Correlations between intestinal fatty acid binding protein and clinical variables in patients

	I-FABP	
Variables	Correlation coefficient	*P*
PASI score	0.18	0.20
Age	0.31	0.02*
Disease duration	0.29	0.03*

*Statistically significant difference (P<0.05). PASI: Psoriasis area severity index, I-FABP: Intestinal fatty acid binding protein

Table 5: Relation of serum levels of intestinal fatty acid binding protein with demographic and clinical data in patients with psoriasis

	I-FABP (ng/mL)		*P*
	Male	Female	
Range	0.10–51.10	1.10–55.30	0.204
Mean±SD	9.58±11.17	10.90±16.50	
Median	6.00	4.80	
IQR	2.80	3.65	
	I-FABP (ng/mL)		***P***
	Plaque type	**Nonplaque type**	
Range	0.10–55.00	3.60–48.40	0.278
Mean±SD	9.66±12.98	14.44±19.04	
Median	5.60	6.30	
IQR	3.25	23.35	
	I-FABP (ng/mL)		***P***
	Skin phototype III	**Skin phototype IV**	
Range	1.10–46.00	0.10–55.30	0.040*
Mean±SD	6.55±9.28	12.74±15.56	
Median	4.80	6.30	
IQR	2.70	6.95	
	I-FABP (ng/mL)		***P***
	Positive family history	**Negative family history**	
Range	3.50–34.50	0.10–55.30	0.567
Mean±SD	7.72±7.59	11.17±15.38	
Median	6.20	5.60	
IQR	2.60	4.30	

*Statistically significant difference (P<0.05). IQR: Interquartile range, I-FABP: Intestinal fatty acid binding protein

Discussion

Psoriasis is a multifactorial, chronic, noninfectious, inflammatory systemic disease. The pathogenesis of psoriasis is complex and not yet fully understood.[13] There may be continuous cross-talk and a mutualistic relationship between intestinal microbiota, the gut barrier, and the immune system. The properly functioning intestinal barrier regulates the translocation of bacteria, endotoxins, and metabolic products into the blood.[14] When this barrier becomes impaired, there may be systemic translocation of harmful products with consequent systemic inflammatory responses. Therefore, enteric dysbiosis and any factor which may compromise intestinal barrier integrity and increase gut permeability could be of importance in the etiopathogenesis of chronic inflammatory diseases, including psoriasis.[15]

I-FABP is a low molecular weight cytoplasmic protein found only in enterocytes, and its increasing blood levels indicate intestinal epithelial cell injury with a consequent dysfunctional intestinal barrier.[16] The present study was designed to evaluate the relationship between serum I-FABP levels and psoriasis. While there is a correlation, the study was not designed to assess which came first, the psoriasis or the defective barrier, or if they both occurred simultaneously from a common stimulant. The lack of correlation between psoriasis severity and I-FABP does not exclude the possibility that impaired intestinal integrity is a trigger for psoriasis development.[17]

The serum level of I-FABP was higher in patients with skin phototype IV than in those with skin phototype III ($P = 0.040$). This finding is in support of the findings denoting the higher frequency of intestinal dysbiosis in people with darker skin.[18] Just as we cannot conclude that intestinal dysbiosis causes dark skin, we cannot conclude that intestinal dysbiosis causes psoriasis.

Conclusion

Small samples size is a limitation of our study; strict exclusion criteria and the pandemic situations were the main responsible factors, also it would be interesting to see if I-FABP decreased after successful treatment of psoriasis with a biologic or other systemic agent that does not directly target gut permeability.

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Conflicts of interest

There are no conflicts of interest.

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